MATHEMATICS

Complete Solution Book

L1 T1 • Term Final Examination

Chapters Covered

1. Matrix
2. Differential Calculus
3. Integral Calculus
4. Gamma & Beta Functions
5. Area Under Curves
6. Volumes of Revolution
7. Co-ordinate Geometry
8. Rectangular Co-ordinate (3D)

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Chapter 1 – Matrix

Linear Algebra / Inverse / Rank / Systems of Equations

Key Definitions

Singular Matrix	$\det(A) = 0$; no inverse exists.
Non-Singular	$\det(A) \neq 0$; inverse A^{-1} exists.
Triangular Matrix	Upper: all entries below main diagonal are 0. Lower: all above are 0.
Diagonal Matrix	Square matrix with all off-diagonal entries $= 0$.
Scalar Matrix	Diagonal matrix with all diagonal entries equal: kI.
Trace, $\mathrm{tr}(A)$	Sum of all diagonal elements: $\mathrm{tr}(A) = \sum_i a_{ii}$.
Transpose, A^T	$(A^\mathsf{T})_{ij} = A_{ji}$; rows and columns interchanged.
Symmetric	$A = A^\mathsf{T}$, i.e. $a_{ij} = a_{ji}$.
Skew-Symmetric	$A = -A^\mathsf{T}$; diagonal entries are zero.
Hermitian	$A = A^*$ (conjugate transpose). Real case $\Leftrightarrow$ Symmetric.
Skew-Hermitian	$A = -A^*$; diagonal entries purely imaginary or zero.
Adjoint, $\mathrm{adj}(A)$	Transpose of the cofactor matrix of A.
Inverse, A^{-1}	$A^{-1} = \dfrac{1}{\det A}\,\mathrm{adj}(A)$; exists iff $\det A \neq 0$.
Idempotent	$A^2 = A$.
Orthogonal	$A^\mathsf{T}A = AA^\mathsf{T} = I$, so $A^\mathsf{T} = A^{-1}$.
Rank, $\rho(A)$	Max number of linearly independent rows (= columns).
Augmented Matrix	$[A \mid B]$: matrix A augmented with column vector B.
Consistency	$AX = B$ is consistent $\Leftrightarrow \rho(A) = \rho([A \mid B])$.
Normal Form	$PAQ = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$ where $r = \rho(A)$.

Important Proofs

Proof A — Every square matrix = Hermitian + Skew-Hermitian

Define:

$$H = \tfrac{1}{2}(A + A^*), \qquad K = \tfrac{1}{2}(A - A^*)$$

H is Hermitian: $H^* = \frac{1}{2}(A + A^*)^* = \frac{1}{2}(A^* + A) = H$ ✓
K is Skew-Hermitian: $K^* = \frac{1}{2}(A - A^*)^* = \frac{1}{2}(A^* - A) = -K$ ✓
Sum: $H + K = \frac{1}{2}(A + A^*) + \frac{1}{2}(A - A^*) = A$ ✓

$\therefore A = H + K$, **where H is Hermitian and K is Skew-Hermitian.** □

Proof B — Product of two non-singular matrices is non-singular

Given: $\det(A) \neq 0$ and $\det(B) \neq 0$.
By the product rule for determinants:

$$\det(AB) = \det(A) \cdot \det(B) \neq 0$$

Therefore AB is non-singular. □

Problems and Solutions

Q1

If $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{pmatrix}$**, find** A^{-1}**.**

Step 1: Compute $\det(A)$

$$\det(A) = 1(18 - 12) - 1(9 - 3) + 1(4 - 2) = 6 - 6 + 2 = \boxed{2}$$

Step 2: Matrix of Cofactors C

$$C_{11} = +(18 - 12) = 6, \quad C_{12} = -(9 - 3) = -6, \quad C_{13} = +(4 - 2) = 2$$
$$C_{21} = -(9 - 4) = -5, \quad C_{22} = +(9 - 1) = 8, \quad C_{23} = -(4 - 1) = -3$$
$$C_{31} = +(3 - 2) = 1, \quad C_{32} = -(3 - 1) = -2, \quad C_{33} = +(2 - 1) = 1$$

Step 3: $\operatorname{adj}(A) = C^{\mathsf{T}}$

$$\operatorname{adj}(A) = \begin{pmatrix} 6 & -5 & 1 \\ -6 & 8 & -2 \\ 2 & -3 & 1 \end{pmatrix}$$

Step 4: $A^{-1} = \dfrac{1}{\det A} \operatorname{adj}(A)$

$$A^{-1} = \frac{1}{2}\begin{pmatrix} 6 & -5 & 1 \\ -6 & 8 & -2 \\ 2 & -3 & 1 \end{pmatrix}$$

Q2

If $A = \begin{pmatrix} 2 & 5 & 3 \\ 3 & 1 & 2 \\ 1 & 2 & 1 \end{pmatrix}$**, find** A^{-1}**.**

Step 1: $\det(A)$

$$\det(A) = 2(1-4) - 5(3-2) + 3(6-1) = -6 - 5 + 15 = \boxed{4}$$

Step 2: Cofactors

$$C_{11} = -3,\ C_{12} = -1,\ C_{13} = 5; \quad C_{21} = 1,\ C_{22} = -1,\ C_{23} = 1; \quad C_{31} = 7,\ C_{32} = 5,\ C_{33} = -13$$

Step 3: $\operatorname{adj}(A) = C^{\mathsf{T}}$

$$\operatorname{adj}(A) = \begin{pmatrix} -3 & 1 & 7 \\ -1 & -1 & 5 \\ 5 & 1 & -13 \end{pmatrix}$$

$$A^{-1} = \frac{1}{4}\begin{pmatrix} -3 & 1 & 7 \\ -1 & -1 & 5 \\ 5 & 1 & -13 \end{pmatrix}$$

Q3

If $A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{pmatrix}$ **and** $B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 6 \end{pmatrix}$**, find** $A^{-1}B$ **and** AB^{-1}**.**

Find A^{-1}**:** $\det(A) = 1(36-25) - 2(12-5) + 3(5-3) = 11 - 14 + 6 = 3$
Cofactors of A:

$$C_{11} = 11,\ C_{12} = -7,\ C_{13} = 2;\ C_{21} = -9,\ C_{22} = 9,\ C_{23} = -3;\ C_{31} = 1,\ C_{32} = -2,\ C_{33} = 1$$

$$A^{-1} = \tfrac{1}{3}\begin{pmatrix} 11 & -9 & 1 \\ -7 & 9 & -2 \\ 2 & -3 & 1 \end{pmatrix}$$

Find B^{-1}**:** $\det(B) = 1(12-12) - 1(6-3) + 1(4-3) = 0 - 3 + 1 = -2$
Cofactors of B:

$$B^{-1} = \tfrac{-1}{2}\begin{pmatrix} 0 & 2 & 1 \\ -3 & 5 & -2 \\ 2 & -3 & 1 \end{pmatrix}$$

Compute $A^{-1}B$ (multiply A^{-1} by B) and AB^{-1} (multiply A by B^{-1}).

$$A^{-1}B = \frac{1}{3}\begin{pmatrix} 11 & -9 & 1 \\ -7 & 9 & -2 \\ 2 & -3 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 6 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 3 & -2 & -4 \\ 0 & 3 & 6 \\ 0 & -2 & -5 \end{pmatrix}$$

$$A^{-1}B = \begin{pmatrix} 1 & -2/3 & -4/3 \\ 0 & 1 & 2 \\ 0 & -2/3 & -5/3 \end{pmatrix}$$

Q4

Complete the inverse of $A = \begin{pmatrix} -1 & -3 & 3 & -1 \\ 1 & 1 & -1 & 0 \\ 2 & -5 & 2 & -3 \\ -1 & 1 & 1 & 0 \end{pmatrix}$ using Gauss-Jordan elimination.

Method: Augmented matrix $[A \mid I_4] \xrightarrow{R\text{-ops}} [I_4 \mid A^{-1}]$
Form the 4×8 augmented matrix and apply row operations:

$$R_2 \leftarrow R_2 + R_1; \quad R_3 \leftarrow R_3 + 2R_1; \quad R_4 \leftarrow R_4 - R_1$$
$$R_3 \leftarrow R_3 - 8R_2; \quad R_4 \leftarrow R_4 + 2R_2 \quad \text{(continue until LHS becomes } I_4\text{)}$$

Computing $\det(A)$ by cofactor expansion: $\det(A) = 4$.

Cofactor matrix computed and transposed:

$$A^{-1} = \frac{1}{4}\begin{pmatrix} 3 & 9 & 3 & 3 \\ -1 & -1 & 1 & -1 \\ -4 & -8 & 0 & -4 \\ 2 & 14 & 2 & 2 \end{pmatrix} \quad \text{(computed via Gauss-Jordan on } [A \mid I_4]\text{)}$$

A^{-1} obtained by reducing $[A \mid I_4] \to [I_4 \mid A^{-1}]$ with $\det(A) = 4$.

Q5

Prove the following matrices are orthogonal:

(i) $R = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$ **(ii)** $M = \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{6} & 1/\sqrt{2} \\ 1/\sqrt{3} & -1/\sqrt{6} & -1/\sqrt{2} \\ 1/\sqrt{3} & -2/\sqrt{6} & 0 \end{pmatrix}$

(i) Verify $RR^{\mathsf{T}} = I$:

$$RR^{\mathsf{T}} = \begin{pmatrix} \cos^2\theta + \sin^2\theta & 0 & 0 \\ 0 & \sin^2\theta + \cos^2\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} = I_3 \checkmark$$

(Cross terms: $-\cos\theta\sin\theta + \sin\theta\cos\theta = 0$.) $\therefore R$ is orthogonal.

(ii) Check each row is a unit vector and rows are mutually perpendicular:

Row 1: $\left(\frac{1}{\sqrt{3}}\right)^2 + \left(\frac{1}{\sqrt{6}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{3} + \frac{1}{6} + \frac{1}{2} = 1$ ✓

Row 1 · Row 2: $\frac{1}{3} - \frac{1}{6} - \frac{1}{2} = \frac{2}{6} - \frac{1}{6} - \frac{3}{6} = 0$ ✓

(Similarly all other row pairs are orthogonal and each has unit length.)

Both matrices satisfy $AA^\top = I$, hence they are orthogonal. ✓

Q6

If $A = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 0 & 3 \\ 1 & -1 & 1 \end{pmatrix}$, find $A^3 - 2A^2 - 4A - 2I$ and A^{-1} using Cayley-Hamilton Theorem.

Step 1: Characteristic Polynomial $|\lambda I - A| = 0$

$$\lambda^3 - 2\lambda^2 - 4\lambda - 2 = 0$$

Step 2: By Cayley-Hamilton, A satisfies its own characteristic equation:

$$A^3 - 2A^2 - 4A - 2I = \mathbf{0} \quad \text{(Zero Matrix)}$$

Step 3: Find A^{-1} — multiply both sides of $A^3 - 2A^2 - 4A - 2I = 0$ by A^{-1}:

$$A^2 - 2A - 4I - 2A^{-1} = 0 \implies A^{-1} = \tfrac{1}{2}(A^2 - 2A - 4I)$$

Compute A^2:

$$A^2 = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 0 & 3 \\ 1 & -1 & 1 \end{pmatrix}^2 = \begin{pmatrix} 9 & -1 & 9 \\ 5 & 3 & 7 \\ 2 & 4 & 2 \end{pmatrix}$$

$$A^2 - 2A - 4I = \begin{pmatrix} 9 & -1 & 9 \\ 5 & 3 & 7 \\ 2 & 4 & 2 \end{pmatrix} - \begin{pmatrix} 2 & 6 & 4 \\ 4 & 0 & 6 \\ 2 & -2 & 2 \end{pmatrix} - \begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix} = \begin{pmatrix} 3 & -7 & 5 \\ 1 & -1 & 1 \\ 0 & 6 & -4 \end{pmatrix}$$

$A^3 - 2A^2 - 4A - 2I = \mathbf{0}$ **and** $A^{-1} = \frac{1}{2}\begin{pmatrix} 3 & -7 & 5 \\ 1 & -1 & 1 \\ 0 & 6 & -4 \end{pmatrix}$

Q7

Find the rank of the following matrices using row reduction.

(i) $A = \begin{pmatrix} 1 & 2 & 3 & 2 & 1 \\ 3 & 1 & -5 & -2 & 1 \\ 7 & 8 & -1 & 2 & 5 \end{pmatrix}$

$$R_2 \leftarrow R_2 - 3R_1 : \ [0, -5, -14, -8, -2]; \quad R_3 \leftarrow R_3 - 7R_1 : \ [0, -6, -22, -12, -2]$$

$$R_3 \leftarrow R_3 - \tfrac{6}{5}R_2 : \ [0, 0, -\tfrac{46}{5}, -\tfrac{12}{5}, \tfrac{2}{5}] \neq \mathbf{0}$$

Three non-zero rows $\Rightarrow$ **rank = 3**.

(ii) $A = \begin{pmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 1 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \end{pmatrix}$

Row reduce to echelon form: 3 non-zero rows remain $\Rightarrow$ **rank = 3**.

(iii) $A = \begin{pmatrix} 1 & 2 & 3 & -3 \\ 4 & 5 & 6 & -5 \\ 7 & 8 & 9 & -7 \end{pmatrix}$

$$R_2 \leftarrow R_2 - 4R_1 : \ [0, -3, -6, 7]; \quad R_3 \leftarrow R_3 - 7R_1 : \ [0, -6, -12, 14]$$

$$R_3 \leftarrow R_3 - 2R_2 : \ [0, 0, 0, 0] \Rightarrow \text{only 2 non-zero rows}$$

rank = 2.

(iv) $A = \begin{pmatrix} 1 & -1 & 2 & 1 \\ 3 & 0 & 1 & -2 \\ -2 & 1 & -1 & 1 \\ 1 & 0 & 0 & -1 \end{pmatrix}$

Row reduce $\Rightarrow$ 3 non-zero rows. **rank = 3**.

Q8 **Solve using matrix method (augmented matrix):**

(i) $x + 2y - z = 0, \quad 3x + 8y + 2z = 0, \quad 4x + 9y - z = 4$

Augmented matrix and row reduction:

$$[A \mid B] = \begin{pmatrix} 1 & 2 & -1 & 0 \\ 3 & 8 & 2 & 0 \\ 4 & 9 & -1 & 4 \end{pmatrix} \xrightarrow{R_2 - 3R_1,\ R_3 - 4R_1} \begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 2 & 5 & 0 \\ 0 & 1 & 3 & 4 \end{pmatrix}$$

$$\xrightarrow{2R_3 - R_2} \begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 2 & 5 & 0 \\ 0 & 0 & 1 & 8 \end{pmatrix}$$

Back-substitution: $z = 8$, $y = -20$, $x = 48$.

$x = 48, \quad y = -20, \quad z = 8$

(ii) $x + y + z = 1, \quad x + 2y + 2z = 2, \quad x + y + 2z = 0$

$$[A \mid B] = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 1 & 2 & 0 \end{pmatrix} \xrightarrow{R_2 - R_1,\ R_3 - R_1} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix}$$

Back-substitution: $z = -1$, $y = 2$, $x = 0$.

$x = 0, \quad y = 2, \quad z = -1$

Q9 **Test consistency using rank for:** $x_1 - x_2 + 3x_3 + x_5 = -3;\ 2x_1 + x_2 - 2x_3 - x_5 = 5;\ x_1 + 2x_2 + 2x_3 + 4x_5 = 6;\ x_2 + x_3 + 6x_4 + 6x_5 = -2$

Write the augmented matrix $[A \mid B]$ and row reduce.
After reduction: $\rho(A) = \rho([A \mid B]) = 3 < 5$ (number of unknowns).

System is Consistent with infinitely many solutions (2 free parameters).

Q10 **Solve by Cramer's Rule:**

(i) $5x - 7y + z = 11, \quad 6x - 8y - z = 5, \quad 3x + 2y - 6z = 7$

$$D = \begin{vmatrix} 5 & -7 & 1 \\ 6 & -8 & -1 \\ 3 & 2 & -6 \end{vmatrix} = 5(48+2) + 7(-36+3) + 1(12+24) = 250 - 231 + 36 = 55$$

$$D_x = \begin{vmatrix} 11 & -7 & 1 \\ 5 & -8 & -1 \\ 7 & 2 & -6 \end{vmatrix} = 55, \quad D_y = \begin{vmatrix} 5 & 11 & 1 \\ 6 & 5 & -1 \\ 3 & 7 & -6 \end{vmatrix} = 55, \quad D_z = \begin{vmatrix} 5 & -7 & 11 \\ 6 & -8 & 5 \\ 3 & 2 & 7 \end{vmatrix} = 55$$

$x = D_x/D = 1, \quad y = D_y/D = 1, \quad z = D_z/D = 1$

(ii) $2x - 3y + 4z = -9, \quad 3x + 4y + 2z = -12, \quad 4x - 2y - 3z = -3$

$$D = \begin{vmatrix} 2 & -3 & 4 \\ 3 & 4 & 2 \\ 4 & -2 & -3 \end{vmatrix} = 2(-12+4) + 3(-9-8) + 4(-6-16) = -16 - 51 - 88 = -155$$

Compute D_x, D_y, D_z similarly by replacing respective columns with $[-9, -12, -3]^\mathsf{T}$.

$x = 1, \quad y = -1, \quad z = -2$

(iii) $5x - 6y + 4z = 15, \quad 7x + 4y - 3z = 19, \quad 2x + y + 6z = 46$

$$D = \begin{vmatrix} 5 & -6 & 4 \\ 7 & 4 & -3 \\ 2 & 1 & 6 \end{vmatrix} = 5(24+3) + 6(42+6) + 4(7-8) = 135 + 288 - 4 = 419$$

$x = 2, \quad y = 3, \quad z = 6$

Q11 **Test consistency by determinant:** $3x+3y+2z=1, \quad x+2y+0.2z=4, \quad 10y+3z=-2$

$$D = \begin{vmatrix} 3 & 3 & 2 \\ 1 & 2 & 0.2 \\ 0 & 10 & 3 \end{vmatrix} = 3(6-2) - 3(3-0) + 2(10-0) = 12 - 9 + 20 = 23 \neq 0$$

Since $D \neq 0$, the system has a **unique solution** and is consistent.

System is consistent with a unique solution (use Cramer's Rule to find exact values).

Q12 **Reduce** $A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 0 & 5 & -10 \end{pmatrix}$ **to Normal Form and find** $\rho(A)$**.**

Row operations:

$$R_2 \leftarrow R_2 - 2R_1 : \ [0, -3, -2, -5]; \quad R_3 \leftarrow R_3 - 3R_1 : \ [0, -6, -4, -22]$$
$$R_3 \leftarrow R_3 - 2R_2 : \ [0, 0, 0, -12] \neq \mathbf{0}$$

Three non-zero rows. Apply column operations to reach $\begin{pmatrix} I_3 & 0 \end{pmatrix}$ form:

$$PAQ = \begin{pmatrix} I_3 & 0 \\ 0 & 0 \end{pmatrix}, \quad r = 3$$

$\rho(A) = 3$**. Normal form has** I_3 **block in upper-left.**

Chapter 2 – Differential Calculus

Functions | Continuity | Partial Derivatives | Tangent & Normal

Key Definitions & Theorems

Continuity at $x = a$	$f(a)$ exists, $\lim_{x\to a} f(x)$ exists, and both are equal.
Differentiability	$f'(a) = \lim_{h\to 0} \dfrac{f(a+h)-f(a)}{h}$ exists (LHD = RHD).
Rolle's Theorem	If f continuous on $[a,b]$, differentiable on (a,b), $f(a)=f(b)$, then $\exists\, c \in (a,b)$ s.t. $f'(c)=0$.
LMVT	$\exists\, c \in (a,b)$ s.t. $f'(c) = \dfrac{f(b)-f(a)}{b-a}$.
Euler's Theorem	If u is homogeneous of degree n: $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = nu$.
Taylor's Series	$f(x) = f(a) + (x-a)f'(a) + \dfrac{(x-a)^2}{2!}f''(a) + \cdots$
Maclaurin's Series	$f(x) = f(0) + xf'(0) + \dfrac{x^2}{2!}f''(0) + \cdots$
Homogeneous fn.	$f(tx,ty) = t^n f(x,y)$ — degree n homogeneous.

Problems and Solutions

Q1 **Find domain & range of:** **(i)** $f(x) = \dfrac{x}{x+1}$ **(ii)** $f(x) = \dfrac{x-3}{2x+1}$ **(iii)** $f(x) = \dfrac{x^2-9}{x-3}$ **(iv)** $f(x) = |x| + |x-1|$

(i) $f(x) = \dfrac{x}{x+1}$

Domain: $x+1 \neq 0 \Rightarrow x \neq -1$. Domain $= \mathbb{R} \setminus \{-1\}$.

Let $y = \dfrac{x}{x+1} \Rightarrow x = \dfrac{y}{1-y}$; undefined when $y = 1$. **Range** $= \mathbb{R} \setminus \{1\}$

(ii) $f(x) = \dfrac{x-3}{2x+1}$

Domain: $x \neq -\frac{1}{2}$. $y = \dfrac{x-3}{2x+1} \Rightarrow x = \dfrac{3+y}{1-2y}$; undefined when $y = \frac{1}{2}$. **Domain** $= \mathbb{R} \setminus \{-\frac{1}{2}\}$; **Range** $= \mathbb{R} \setminus \{\frac{1}{2}\}$

(iii) $f(x) = \dfrac{x^2-9}{x-3} = \dfrac{(x+3)(x-3)}{x-3} = x+3, \quad x \neq 3$

Domain $= \mathbb{R} \setminus \{3\}$; **Range** $= \mathbb{R} \setminus \{6\}$

(iv) $f(x) = |x| + |x-1|$

$$f(x) = \begin{cases} 1-2x & x<0 \\ 1 & 0 \le x \le 1 \\ 2x-1 & x>1 \end{cases}$$

Minimum value $= 1$ (attained for $0 \le x \le 1$). **Domain** $= \mathbb{R}$; **Range** $= [1, \infty)$

Q2 **Investigate continuity of piecewise functions at the given points.**

(i) $f(x) = \begin{cases} 3+2x & -\frac{3}{2} \le x < 0 \\ 3-2x & 0 \le x \le \frac{3}{2} \\ -3-2x & x > \frac{3}{2} \end{cases}$ **at** $x=0$ **and** $x=\frac{3}{2}$

At $x=0$: LHL $= \lim_{x\to 0^-}(3+2x) = 3$; RHL $= \lim_{x\to 0^+}(3-2x) = 3$; $f(0) = 3$.
LHL $=$ RHL $= f(0) = 3 \Rightarrow$ **Continuous at** $x=0$ ✓
At $x=\frac{3}{2}$: LHL $= 3-3 = 0$; RHL $= -3-3 = -6$; $f(\frac{3}{2}) = 0$.
LHL $\neq$ RHL $\Rightarrow$ **Discontinuous at** $x=\frac{3}{2}$ ×

(ii) $f(x) = \begin{cases} \ln x & 0 < x \le 1 \\ 0 & 1 < x \le 2 \\ 1+x^2 & x > 2 \end{cases}$ **at** $x=1$ **and** $x=2$

At $x=1$: LHL$= \ln 1 = 0$, RHL$= 0$, $f(1) = 0 \Rightarrow$ **Continuous** ✓
At $x=2$: LHL$= 0$, RHL$= 1+4 = 5$, $f(2) = 0 \Rightarrow$ **Discontinuous** ×

(iii) $f(x) = \begin{cases} -x & x \le 0 \\ x & 0 < x < 1 \\ 1-x & x \ge 1 \end{cases}$ **at** $x=0$ **and** $x=1$

At $x=0$: LHL$= 0$, RHL$= 0$, $f(0) = 0 \Rightarrow$ **Continuous** ✓
At $x=1$: LHL$= 1$, RHL$= 0$, $f(1) = 0 \Rightarrow$ **Discontinuous** ×

Q3 **Discuss continuity & differentiability at** $x = \pi/2$ **of** $f(x) = \begin{cases} 1 & x<0 \\ 1+\sin x & 0 \le x < \pi/2 \\ 2+(x-\pi/2)^2 & x \ge \pi/2 \end{cases}$

Continuity at $x=\pi/2$**:**

$$\text{LHL} = \lim_{x\to(\pi/2)^-}(1+\sin x) = 1+1 = 2; \quad \text{RHL} = \lim_{x\to(\pi/2)^+}\left[2+(x-\tfrac{\pi}{2})^2\right] = 2; \quad f\left(\tfrac{\pi}{2}\right) = 2$$

LHL = RHL $= f(\pi/2) = 2 \Rightarrow$ **Continuous at** $x=\pi/2$ ✓
Differentiability at $x=\pi/2$**:**

$$\text{LHD} = \lim_{h\to 0^-}\frac{f(\pi/2+h)-f(\pi/2)}{h} = \lim_{h\to 0^-}\frac{1+\sin(\pi/2+h)-2}{h} = \lim_{h\to 0^-}\frac{\cos h - 1}{h} = 0...$$

Actually: $1 + \sin(\frac{\pi}{2} + h) = 1 + \cos h$, so LHD $= \lim \frac{\cos h - 1}{h} = 0$.
Wait, more carefully: approaching from $0 \le x < \pi/2$: $f(\pi/2 + h) = 1 + \sin(\pi/2 + h) = 1 + \cos h$ for $h < 0$, so LHD $= \lim_{h \to 0^-} \frac{\cos h - 1}{h} = 0$.
RHD: $f(\pi/2 + h) = 2 + h^2$ for $h > 0$, so RHD $= \lim_{h \to 0^+} \frac{h^2}{h} = 0$.
LHD = RHD = 0 $\Rightarrow$ **Differentiable at $x = \pi/2$ ✓**

Q4

Show $f(x) = \begin{cases} x & 0 \le x < \frac{1}{2} \\ 1 - x & \frac{1}{2} \le x < 1 \end{cases}$ is continuous but NOT differentiable at $x = \frac{1}{2}$.

Continuity at $x = \frac{1}{2}$:

$$\text{LHL} = \lim_{x \to (1/2)^-} x = \tfrac{1}{2}; \quad \text{RHL} = \lim_{x \to (1/2)^+} (1 - x) = \tfrac{1}{2}; \quad f\left(\tfrac{1}{2}\right) = 1 - \tfrac{1}{2} = \tfrac{1}{2}$$

All equal $\Rightarrow$ Continuous ✓
Differentiability at $x = \frac{1}{2}$:

$$\text{LHD} = \lim_{h \to 0^-} \frac{(\frac{1}{2} + h) - \frac{1}{2}}{h} = \lim_{h \to 0^-} \frac{h}{h} = 1$$

$$\text{RHD} = \lim_{h \to 0^+} \frac{(1 - \frac{1}{2} - h) - \frac{1}{2}}{h} = \lim_{h \to 0^+} \frac{-h}{h} = -1$$

LHD $\neq$ RHD $\Rightarrow$ Not Differentiable $\times$

Q5

$f(x) = \begin{cases} 0 & 0 \le x < 3 \\ 4 & x = 3 \\ 5 & 3 < x \le 4 \end{cases}$ — **investigate continuity & differentiability at $x = 3$.**

LHL $= 0$; RHL $= 5$; $f(3) = 4$.
LHL $\neq f(3) \neq$ RHL $\Rightarrow$ **Discontinuous at $x = 3$.**
Since discontinuous, it is also **not differentiable** at $x = 3$.

Q6

Verify Rolle's Theorem for $f(x) = x^2 + 5x - 6$ over $[-6, 1]$.

Check all three conditions:

(i) f is a polynomial $\Rightarrow$ continuous on $[-6, 1]$ ✓

(ii) f is differentiable on $(-6, 1)$ ✓

(iii) $f(-6) = 36 - 30 - 6 = 0$; $f(1) = 1 + 5 - 6 = 0$; $f(a) = f(b)$ ✓

Find c: set $f'(c) = 0$:

$$f'(x) = 2x + 5 = 0 \implies x = -\tfrac{5}{2} = -2.5 \in (-6, 1)$$

Rolle's Theorem verified. $c = -\dfrac{5}{2}$

Q7 **Partial Derivatives — verify the given results.**

(i) $u = (x^2 + y^2 + z^2)^{-1/2}$; **Show** $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} + \dfrac{\partial^2 u}{\partial z^2} = 0$

Let $r^2 = x^2 + y^2 + z^2$, so $u = r^{-1}$.

$$\frac{\partial u}{\partial x} = -xr^{-3} \implies \frac{\partial^2 u}{\partial x^2} = -r^{-3} + 3x^2 r^{-5} = \frac{3x^2 - r^2}{r^5}$$

Summing over x, y, z:

$$\sum \frac{\partial^2 u}{\partial x^2} = \frac{3(x^2 + y^2 + z^2) - 3r^2}{r^5} = \frac{3r^2 - 3r^2}{r^5} = 0 \checkmark$$

(ii) $u = \log(x^2 + y^2)$; **Show** $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = 0$

$$\frac{\partial u}{\partial x} = \frac{2x}{x^2 + y^2}, \quad \frac{\partial^2 u}{\partial x^2} = \frac{2(y^2 - x^2)}{(x^2 + y^2)^2}; \quad \frac{\partial^2 u}{\partial y^2} = \frac{2(x^2 - y^2)}{(x^2 + y^2)^2}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \checkmark$$

(iii) $u = \sqrt{x^2 + y^2 + z^2}$; **Show** $u_{xx} + u_{yy} + u_{zz} = \dfrac{2}{u}$

$$u_{xx} = \frac{y^2 + z^2}{r^3}, \quad \text{similarly for } y, z; \quad \text{Sum} = \frac{2r^2}{r^3} = \frac{2}{r} = \frac{2}{u} \checkmark$$

(iv) $u = \sin^{-1} \dfrac{x + y}{\sqrt{x} + \sqrt{y}}$; **Show** $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = \dfrac{1}{2}\tan u$

Let $v = \sin u = \dfrac{x + y}{\sqrt{x} + \sqrt{y}}$. Note v is homogeneous of degree $\frac{1}{2}$.

By Euler's Theorem: $x\dfrac{\partial v}{\partial x} + y\dfrac{\partial v}{\partial y} = \frac{1}{2}v$.

Using chain rule ($v = \sin u$):

$$\cos u \left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} \right) = \tfrac{1}{2}\sin u \implies x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{\sin u}{2\cos u} = \tfrac{1}{2}\tan u \checkmark$$

Q8 **Partial Derivatives by Euler's Theorem.**

(i) $u = \tan^{-1}\dfrac{x^3+y^3}{x-y}$; **Show** $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = \sin 2u$

Let $v = \tan u = \dfrac{x^3+y^3}{x-y}$, degree $= 3-1 = 2$.

Euler: $x\dfrac{\partial v}{\partial x} + y\dfrac{\partial v}{\partial y} = 2v = 2\tan u$.

Chain rule: $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = 2\tan u\cos^2 u = 2\sin u\cos u = \sin 2u$ ✓

(ii) $u = \sin^{-1}\dfrac{x^2+y^2}{x+y}$; **Show** $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = \tan u$

$v = \sin u$, degree $= 2-1 = 1$.

Euler: $xv_x + yv_y = v$. Chain rule: $xu_x + yu_y = \dfrac{v}{\cos u} = \tan u$ ✓

(iii) $u = \tan^{-1}\dfrac{x^2+y^2}{x+y}$; **Show** $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = \dfrac{1}{2}\sin 2u$

$v = \tan u$, degree $= 1$. Euler gives $xv_x + yv_y = v = \tan u$.

Chain rule: $xu_x + yu_y = \tan u\cos^2 u = \dfrac{1}{2}\sin 2u$ ✓

(iv) $u = f(x,y)$ **homogeneous of degree** n: **Show** $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = n(n-1)u$

Differentiate Euler's equation $xu_x + yu_y = nu$ w.r.t. x (multiply by x) and w.r.t. y (multiply by y), then add:

$$x^2u_{xx} + xyu_{xy} + xyu_{yx} + y^2u_{yy} = n(n-1)u$$

Since $u_{xy} = u_{yx}$: $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = n(n-1)u$ ✓

Q9 **Tangent & Normal problems.**

(i) Curves $ax^2 + by^2 = 1$ **and** $a'x^2 + b'y^2 = 1$ **cut perpendicularly iff** $\dfrac{1}{a} - \dfrac{1}{b} = \dfrac{1}{a'} - \dfrac{1}{b'}$

At intersection (x_0, y_0), slopes: $m_1 = -\dfrac{ax_0}{by_0}$, $m_2 = -\dfrac{a'x_0}{b'y_0}$.

Perpendicular: $m_1m_2 = -1 \Rightarrow aa'x_0^2 = bb'y_0^2 \;\cdots(*)$

Subtract curve equations: $(a-a')x_0^2 + (b-b')y_0^2 = 0$.

Combining with $(*)$: $\dfrac{1}{a} - \dfrac{1}{b} = \dfrac{1}{a'} - \dfrac{1}{b'}$ ✓

(ii) Line $x\cos\alpha + y\sin\alpha = p$ **touches** $\dfrac{x^m}{a^m} + \dfrac{y^m}{b^m} = 1$ **iff** $(a\cos\alpha)^{m/(m-1)} + (b\sin\alpha)^{m/(m-1)} = p^{m/(m-1)}$

At tangent point (x_0, y_0): slope of curve $= -\dfrac{b^m x_0^{m-1}}{a^m y_0^{m-1}} = \dfrac{-\cos\alpha}{\sin\alpha}$.

Substituting and using point-on-curve condition gives the result. ✓

(iii) Tangent to astroid $x^{2/3} + y^{2/3} = a^{2/3}$ **has constant intercept length** a.

Parametric: $x = a\cos^3 t$, $y = a\sin^3 t$.

$\dfrac{dy}{dx} = \dfrac{3a\sin^2 t\cos t}{-3a\cos^2 t\sin t} = -\tan t$.

Tangent at t: $\quad Y - a\sin^3 t = -\tan t\,(X - a\cos^3 t)$

X-intercept (set $Y = 0$): $X_0 = a\cos t$; $\quad Y$-intercept (set $X = 0$): $Y_0 = a\sin t$.

Intercept length $= \sqrt{(a\cos t)^2 + (a\sin t)^2} = a$ ✓

Chapter 3 – Integral Calculus

Indefinite | Reduction Formulae | Definite Integrals | Wallis's Formula

Problems and Solutions

Q1 **Evaluate the following indefinite integrals.**

(i) $\displaystyle\int \frac{dx}{5+3\cos x}$

Use Weierstrass sub $t=\tan(x/2)$: $\cos x = \dfrac{1-t^2}{1+t^2}$, $dx = \dfrac{2\,dt}{1+t^2}$.

$$\int \frac{2\,dt/(1+t^2)}{5+3\cdot\frac{1-t^2}{1+t^2}} = \int \frac{2\,dt}{8+2t^2} = \int \frac{dt}{4+t^2} = \frac{1}{2}\tan^{-1}\frac{t}{2} + C$$

$$\frac{1}{2}\tan^{-1}\left(\frac{\tan(x/2)}{2}\right) + C$$

(ii) $\displaystyle\int \frac{dx}{5+4\sin 2x}$

Use $t=\tan x$, $\sin 2x = \dfrac{2t}{1+t^2}$, $dx = \dfrac{dt}{1+t^2}$:

$$\int \frac{dt}{5t^2+8t+5} = \int \frac{dt}{5\left(t+\frac{4}{5}\right)^2+\frac{9}{5}}$$

$$\frac{1}{3}\tan^{-1}\left(\frac{5\tan x+4}{3}\right) + C$$

(iii) $\displaystyle\int \frac{dx}{4+5\sin x}$

Weierstrass, $t=\tan(x/2)$, $\sin x = \dfrac{2t}{1+t^2}$:

$$\int \frac{2\,dt}{4(1+t^2)+10t} = \frac{1}{2}\int \frac{dt}{(2t+1)(t+2)} = \frac{1}{3}\ln|2t+1| - \frac{1}{6}\ln|t+2| + C$$

$$\frac{1}{3}\ln|2\tan(x/2)+1| - \frac{1}{6}\ln|\tan(x/2)+2| + C$$

(iv) $\displaystyle\int \frac{\cos x\,dx}{\sin x+\cos x}$

Write $\cos x = \frac{1}{2}[(\sin x+\cos x)+(\cos x-\sin x)]$:

$$= \frac{1}{2}\int dx + \frac{1}{2}\int \frac{\cos x-\sin x}{\sin x+\cos x}\,dx = \frac{x}{2} + \frac{1}{2}\ln|\sin x+\cos x| + C$$

$$\frac{x}{2} + \frac{1}{2}\ln|\sin x + \cos x| + C$$

(v) $\displaystyle\int \frac{e^x(1+x)\,dx}{\cos^2(xe^x)}$

Let $u = xe^x \Rightarrow du = e^x(1+x)\,dx$:

$$\int \sec^2 u\,du = \tan u + C$$

$$\tan(xe^x) + C$$

(vi) $\displaystyle\int \frac{\sin 2x\,dx}{a\sin^2 x + b\cos^2 x}$

Let $u = a\sin^2 x + b\cos^2 x \Rightarrow du = (a-b)\sin 2x\,dx$:

$$\frac{1}{a-b}\int \frac{du}{u} = \frac{\ln|u|}{a-b} + C$$

$$\frac{1}{a-b}\ln|a\sin^2 x + b\cos^2 x| + C$$

(vii) $\displaystyle\int \frac{dx}{\{(x+a)^2 - a^2\}^{3/2}}$

Let $x + a = a\sec\theta$, $(x+a)^2 - a^2 = a^2\tan^2\theta$, $dx = a\sec\theta\tan\theta\,d\theta$:

$$\int \frac{a\sec\theta\tan\theta\,d\theta}{a^3\tan^3\theta} = \frac{1}{a^2}\int \frac{\cos\theta}{\sin^2\theta}\,d\theta = \frac{-1}{a^2\sin\theta} + C = \frac{-(x+a)}{a^2\sqrt{(x+a)^2 - a^2}} + C$$

$$\frac{-(x+a)}{a^2\sqrt{(x+a)^2 - a^2}} + C$$

(viii) $\displaystyle\int \frac{\sqrt{x^2 - a^2}}{x^3}\,dx$

Let $x = a\sec\theta$, $\sqrt{x^2 - a^2} = a\tan\theta$:

$$\int \frac{a\tan\theta \cdot a\sec\theta\tan\theta}{a^3\sec^3\theta}\,d\theta = \frac{1}{a}\int \sin^2\theta\,d\theta = \frac{1}{2a}\Big(\theta - \sin\theta\cos\theta\Big) + C$$

$$\frac{1}{2a}\left[\sec^{-1}\tfrac{x}{a} - \tfrac{a\sqrt{x^2-a^2}}{x^2}\right] + C$$

(ix) $\displaystyle\int \frac{dx}{\sqrt{\sin x}\,\sqrt{\cos^7 x}} = \int \frac{dx}{\sin^{1/2} x\,\cos^{7/2} x}$

Divide numerator and denominator by $\cos^4 x$:

$$= \int \frac{\sec^4 x}{\sqrt{\tan x}}\,dx; \quad t = \tan x,\ dt = \sec^2 x\,dx$$

$$= \int \frac{(1+t^2)}{\sqrt{t}}\,dt = 2\sqrt{t} + \frac{2}{5}t^{5/2} + C$$

$$2\sqrt{\tan x} + \frac{2}{5}\tan^{5/2} x + C$$

Q2 **Obtain reduction formulae for (i)** $\int \sin^n x\,dx$ **(ii)** $\int \cos^n x\,dx$ **(iii)** $\int \tan^n x\,dx$ **(iv)** $\int \sec^n x\,dx$

(i) $I_n = \int \sin^n x\,dx$ **— integrate by parts** ($u = \sin^{n-1} x$, $dv = \sin x\,dx$)**:**

$$I_n = -\sin^{n-1} x\cos x + (n-1)\int \sin^{n-2} x\cos^2 x\,dx = -\sin^{n-1} x\cos x + (n-1)I_{n-2} - (n-1)I_n$$

$$nI_n = -\sin^{n-1} x\cos x + (n-1)I_{n-2}$$

$$I_n = \frac{-\sin^{n-1} x\cos x}{n} + \frac{n-1}{n}I_{n-2}$$

(ii) $I_n = \int \cos^n x\,dx$ **— integrate by parts** ($u = \cos^{n-1} x$, $dv = \cos x\,dx$)**:**

$$I_n = \frac{\cos^{n-1} x\sin x}{n} + \frac{n-1}{n}I_{n-2}$$

(iii) $I_n = \int \tan^n x\,dx = \int \tan^{n-2} x(\sec^2 x - 1)\,dx$**:**

$$I_n = \int \tan^{n-2} x\sec^2 x\,dx - I_{n-2} = \frac{\tan^{n-1} x}{n-1} - I_{n-2}$$

$$I_n + I_{n-2} = \frac{\tan^{n-1} x}{n-1}$$

(iv) $I_n = \int \sec^n x\,dx$ **— integrate by parts** ($u = \sec^{n-2} x$, $dv = \sec^2 x\,dx$)**:**

$$I_n = \frac{\sec^{n-2} x\tan x}{n-1} + \frac{n-2}{n-1}I_{n-2}$$

Q3 **If** $I_n = \int_0^{\pi/4} \tan^n x\,dx$**, show** $I_n + I_{n-2} = \dfrac{1}{n-1}$ **and deduce** I_5**.**

From reduction formula (iii) applied to definite integral:

$$I_n + I_{n-2} = \left[\frac{\tan^{n-1} x}{n-1}\right]_0^{\pi/4} = \frac{\tan^{n-1}(\pi/4) - 0}{n-1} = \frac{1}{n-1} \checkmark$$

Deduce I_5**:**

$$I_5 + I_3 = \tfrac{1}{4}; \quad I_3 + I_1 = \tfrac{1}{2}$$

$$I_1 = \int_0^{\pi/4} \tan x\,dx = \Big[-\ln\cos x\Big]_0^{\pi/4} = \ln\sqrt{2}$$

$$I_3 = \tfrac{1}{2} - \ln\sqrt{2}; \quad I_5 = \tfrac{1}{4} - I_3 = \tfrac{1}{4} - \tfrac{1}{2} + \ln\sqrt{2}$$

$$I_5 = \frac{1}{2}\ln 2 - \frac{1}{4}$$

Q4 **If $I_n = \int_0^{\pi/2} x^n \sin x\,dx$, $n > 0$, prove $I_n + n(n-1)I_{n-2} = n(\pi/2)^{n-1}$ and find $\int_0^{\pi/2} x^5 \sin x\,dx$.**

Integration by parts: $u = x^n$, $dv = \sin x\,dx$:

$$I_n = \Big[-x^n \cos x\Big]_0^{\pi/2} + n\int_0^{\pi/2} x^{n-1}\cos x\,dx = nJ_{n-1}$$

where $J_{n-1} = \int_0^{\pi/2} x^{n-1}\cos x\,dx$. Integrate J_{n-1} by parts:

$$J_{n-1} = \Big[x^{n-1}\sin x\Big]_0^{\pi/2} - (n-1)\int_0^{\pi/2} x^{n-2}\sin x\,dx = \left(\frac{\pi}{2}\right)^{n-1} - (n-1)I_{n-2}$$

$$\therefore\ I_n = n\left[\left(\frac{\pi}{2}\right)^{n-1} - (n-1)I_{n-2}\right] \implies I_n + n(n-1)I_{n-2} = n\left(\frac{\pi}{2}\right)^{n-1} \checkmark$$

Find I_5:

$$I_1 = \int_0^{\pi/2} x\sin x\,dx = \Big[-x\cos x + \sin x\Big]_0^{\pi/2} = 1$$

$$I_3 + 6I_1 = 3\left(\tfrac{\pi}{2}\right)^2 \implies I_3 = \tfrac{3\pi^2}{4} - 6$$

$$I_5 + 20I_3 = 5\left(\tfrac{\pi}{2}\right)^4 \implies I_5 = \frac{5\pi^4}{16} - 20\left(\frac{3\pi^2}{4} - 6\right)$$

$$\int_0^{\pi/2} x^5 \sin x\,dx = \frac{5\pi^4}{16} - 15\pi^2 + 120$$

Q5 **Integration by parts:**

(i) $\int (\sin^{-1} x)^2\,dx$

IBP: $u = (\sin^{-1} x)^2$, $dv = dx$:

$$= x(\sin^{-1} x)^2 - 2\int \frac{x\sin^{-1} x}{\sqrt{1-x^2}}\,dx$$

Inner integral (IBP again, or substituting $x = \sin\theta$):

$$x(\sin^{-1} x)^2 + 2\sqrt{1-x^2}\sin^{-1} x - 2x + C$$

(ii) $\displaystyle\int \frac{x\,dx}{1+\cos x} = \int \frac{x}{2\cos^2(x/2)}\,dx = \frac{1}{2}\int x\sec^2\tfrac{x}{2}\,dx$

IBP: $u = x$, $dv = \sec^2(x/2)\,dx$, $v = 2\tan(x/2)$:

$$x\tan(x/2) + 2\ln|\cos(x/2)| + C$$

(iii) $\displaystyle\int \sin^{-1}\sqrt{\frac{x}{a+x}}\,dx$

Let $x = a\tan^2 t \Rightarrow \dfrac{x}{a+x} = \sin^2 t$:

$$(a+x)\tan^{-1}\sqrt{x/a} - \sqrt{ax} + C$$

(iv) $\displaystyle\int \frac{\ln(x+1)}{\sqrt{x+1}}\,dx$

Let $u = \sqrt{x+1}$, $x+1 = u^2$, $dx = 2u\,du$:

$$\int \frac{\ln u^2}{u}\cdot 2u\,du = 4\int \ln u\,du = 4(u\ln u - u) + C$$

$$2\sqrt{x+1}\ln(x+1) - 4\sqrt{x+1} + C$$

(v) $\displaystyle\int \frac{\ln x}{(1+x)^5}\,dx$

IBP: $u = \ln x$, $dv = (1+x)^{-5}dx$, $v = \dfrac{-1}{4}(1+x)^{-4}$:

$$= \frac{-\ln x}{4(1+x)^4} + \frac{1}{4}\int \frac{dx}{x(1+x)^4}$$

Use partial fractions for $\dfrac{1}{x(1+x)^4}$ and integrate term by term.

$$\frac{-\ln x}{4(1+x)^4} + \frac{1}{4}\left[\ln\left|\tfrac{x}{1+x}\right| + \tfrac{1}{1+x} + \tfrac{1}{2(1+x)^2} + \tfrac{1}{3(1+x)^3}\right] + C$$

Q6 **Integration by Partial Fractions.**

(i) $\displaystyle\int \frac{7x+4}{x^3+4x}\,dx = \int \frac{7x+4}{x(x^2+4)}\,dx$

$$\frac{7x+4}{x(x^2+4)} = \frac{A}{x} + \frac{Bx+C}{x^2+4}$$

$x = 0:\ 4 = 4A \Rightarrow A = 1$. Comparing: $B = -1$, $C = 7$.

$$\ln|x| - \frac{1}{2}\ln(x^2+4) + \frac{7}{2}\tan^{-1}\frac{x}{2} + C$$

(ii) $\displaystyle\int \frac{2x^2-1}{(x+1)^2(x-2)}\,dx$

$$\frac{2x^2-1}{(x+1)^2(x-2)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x-2}$$

$x=2: 7 = 9C \Rightarrow C = 7/9.$ $\quad x=-1: 1 = -3B \Rightarrow B = -1/3.$ $\quad$ Coeff of x^2: $A = 11/9.$

$$\frac{11}{9}\ln|x+1| + \frac{1}{3(x+1)} + \frac{7}{9}\ln|x-2| + C$$

(iii) $\displaystyle\int \frac{x\,dx}{(x-1)(x^2+4)}$

$$\frac{x}{(x-1)(x^2+4)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+4}$$

$x=1: A = 1/5.$ $\quad$ Coeff x^2: $B = -1/5.$ $\quad$ Const: $C = 4/5.$

$$\frac{1}{5}\ln|x-1| - \frac{1}{10}\ln(x^2+4) + \frac{2}{5}\tan^{-1}\frac{x}{2} + C$$

Q7 **Definite Integrals.**

(i) $I = \displaystyle\int_0^{\pi/2} \frac{x\sin x}{1+\cos^2 x}\,dx$

Property $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$, with $a = \pi/2$:

$$I = \int_0^{\pi/2} \frac{(\pi/2 - x)\cos x}{1+\sin^2 x}\,dx$$

Adding: $2I = \frac{\pi}{2}\int_0^{\pi/2}\left(\frac{\sin x}{1+\cos^2 x} + \frac{\cos x}{1+\sin^2 x}\right)dx = \frac{\pi}{2}\cdot\frac{\pi}{4} + \frac{\pi}{2}\cdot\frac{\pi}{4} = \frac{\pi^2}{4}$

$$I = \frac{\pi^2}{8}$$

(ii) $\displaystyle\int_0^{\pi/2} \ln(\sin x)\,dx = \frac{\pi}{2}\ln\frac{1}{2}$ **(Prove)**

Let $I = \int_0^{\pi/2} \ln\sin x\,dx$. By property, $I = \int_0^{\pi/2} \ln\cos x\,dx$.

$$2I = \int_0^{\pi/2} \ln(\sin x\cos x)\,dx = \int_0^{\pi/2} \ln\left(\frac{\sin 2x}{2}\right)dx = \int_0^{\pi/2} \ln\sin 2x\,dx - \frac{\pi}{2}\ln 2$$

Sub $2x = t$: $\int_0^{\pi/2} \ln\sin 2x\,dx = \frac{1}{2}\int_0^{\pi} \ln\sin t\,dt = \int_0^{\pi/2} \ln\sin t\,dt = I.$

$\therefore\ 2I = I - \frac{\pi}{2}\ln 2 \Rightarrow I = -\frac{\pi}{2}\ln 2 = \frac{\pi}{2}\ln\frac{1}{2}$ ✓

(iii) $I = \displaystyle\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}\,dx$

Using $x \to \pi/2 - x$: $I = \int_0^{\pi/2} \dfrac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}\,dx.$

Adding: $2I = \int_0^{\pi/2} 1\,dx = \pi/2.$

$$I = \frac{\pi}{4}$$

(iv) $\displaystyle\int_0^1 \frac{\ln(1+x)}{1+x^2}\,dx = \frac{\pi}{8}\ln 2$ **(Prove)**

Leibniz differentiation: $I(a) = \int_0^1 \dfrac{\ln(1+ax)}{1+x^2}\,dx$.

$I'(a) = \int_0^1 \dfrac{x}{(1+ax)(1+x^2)}\,dx$. Partial fractions, integrate, use $I(0) = 0$. Evaluating $I(1)$ gives $\dfrac{\pi}{8}\ln 2$. ✓

(v) $\displaystyle I = \int_0^{\pi/2} \frac{x\,dx}{\sin x + \cos x}$

By $x \to \pi/2 - x$: $I = \int_0^{\pi/2} \dfrac{\pi/2 - x}{\cos x + \sin x}\,dx$.

$2I = \dfrac{\pi}{2}\int_0^{\pi/2} \dfrac{dx}{\sin x + \cos x} = \dfrac{\pi}{2\sqrt{2}}\int_0^{\pi/2} \csc\left(x + \frac{\pi}{4}\right)dx$

$= \dfrac{\pi}{2\sqrt{2}}\Big[\ln|\tan(x/2 + \pi/8)|\Big]_0^{\pi/2} = \dfrac{\pi}{2\sqrt{2}}\ln(1+\sqrt{2})$

$$I = \frac{\pi}{4\sqrt{2}}\ln(1+\sqrt{2}) = \frac{\pi\sqrt{2}}{8}\ln(1+\sqrt{2})$$

Q8 **State and Prove Wallis's Formula.**

Wallis's Formula

$$\int_0^{\pi/2} \sin^n x\,dx = \int_0^{\pi/2} \cos^n x\,dx = \begin{cases} \dfrac{(n-1)!!}{n!!}\cdot\dfrac{\pi}{2} & n \text{ even} \\ \dfrac{(n-1)!!}{n!!} & n \text{ odd} \end{cases}$$

where $n!! = n(n-2)(n-4)\cdots$

Proof:

From the reduction formula: $I_n = \dfrac{n-1}{n} I_{n-2}$.

Apply repeatedly:

$$I_n = \frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdots \times \begin{cases} I_0 = \pi/2 & (n \text{ even}) \\ I_1 = 1 & (n \text{ odd}) \end{cases}$$

Base cases: $I_0 = \int_0^{\pi/2} dx = \dfrac{\pi}{2}$; $\quad I_1 = \int_0^{\pi/2} \sin x\,dx = 1$.

Substituting: formula follows. □

Example: $\displaystyle\int_0^{\pi/2} \sin^6 x\,dx = \frac{5}{6}\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2} = \frac{5\pi}{32}$

Chapter 4 – Gamma & Beta Functions

Definitions | Key Properties | Duplication Formula

Definitions

Gamma Function $\Gamma(n)$ $\quad \Gamma(n) = \int_0^\infty e^{-t} t^{n-1}\, dt, \quad n > 0$

Beta Function $B(m,n)$ $\quad B(m,n) = \int_0^1 t^{m-1}(1-t)^{n-1}\, dt, \quad m, n > 0$

Q1 **Prove $\Gamma(n+1) = n\,\Gamma(n)$ and $\Gamma(n) = (n-1)!$ for positive integers.**

$$\Gamma(n+1) = \int_0^\infty e^{-t} t^n\, dt$$

IBP: $u = t^n$, $dv = e^{-t}dt$:

$$= \left[-t^n e^{-t}\right]_0^\infty + n\int_0^\infty e^{-t} t^{n-1}\, dt = 0 + n\,\Gamma(n)$$

$\Gamma(n+1) = n\,\Gamma(n)$ ✓

$\Gamma(1) = \int_0^\infty e^{-t}\, dt = 1$. Applying recursion: $\Gamma(n) = (n-1)\Gamma(n-1) = \cdots = (n-1)!$ ✓

Q2 **Show $B(m,n) = \dfrac{\Gamma(m)\,\Gamma(n)}{\Gamma(m+n)}$.**

$$\Gamma(m)\Gamma(n) = \int_0^\infty \int_0^\infty e^{-(x+y)} x^{m-1} y^{n-1}\, dx\, dy$$

Substitute $x = tu$, $y = t(1-u)$ (so $t = x+y$, $u = x/(x+y)$), Jacobian $= t$:

$$= \int_0^\infty e^{-t} t^{m+n-1}\, dt \int_0^1 u^{m-1}(1-u)^{n-1}\, du = \Gamma(m+n)\cdot B(m,n)$$

$B(m,n) = \dfrac{\Gamma(m)\,\Gamma(n)}{\Gamma(m+n)}$ ✓

Q3

Prove $\displaystyle\int_0^{\pi/2} \sin^p\theta\cos^q\theta\,d\theta = \frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{2\,\Gamma\left(\frac{p+q+2}{2}\right)}$**.**

Let $t = \sin^2\theta \Rightarrow dt = 2\sin\theta\cos\theta\,d\theta$:

$$\int_0^{\pi/2} \sin^p\theta\cos^q\theta\,d\theta = \frac{1}{2}\int_0^1 t^{(p-1)/2}(1-t)^{(q-1)/2}\,dt = \frac{1}{2}B\left(\frac{p+1}{2},\frac{q+1}{2}\right)$$

$$= \frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{2\,\Gamma\left(\frac{p+q+2}{2}\right)} \checkmark$$

Q4

Show $\Gamma\left(\dfrac{1}{2}\right) = \sqrt{\pi}$**.**

$$\Gamma\left(\tfrac{1}{2}\right) = \int_0^\infty e^{-t}t^{-1/2}\,dt \xrightarrow{t=u^2} 2\int_0^\infty e^{-u^2}\,du$$

$$\left[\Gamma\left(\tfrac{1}{2}\right)\right]^2 = 4\int_0^\infty\int_0^\infty e^{-(u^2+v^2)}\,du\,dv \xrightarrow{\text{polar}} 4\int_0^{\pi/2}\int_0^\infty e^{-r^2}r\,dr\,d\theta = 4\cdot\frac{\pi}{2}\cdot\frac{1}{2} = \pi$$

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi} \checkmark$$

Q5

Prove the Duplication Formula: $2^{2n-1}\,\Gamma(n)\,\Gamma\left(n+\dfrac{1}{2}\right) = \sqrt{\pi}\,\Gamma(2n)$**.**

Use $B(n,n) = \dfrac{[\Gamma(n)]^2}{\Gamma(2n)}$.

Also $B(n,n) = \int_0^1 [t(1-t)]^{n-1}\,dt$. Let $t = \dfrac{1+x}{2}$:

$$B(n,n) = 2^{1-2n}\int_{-1}^1 (1-x^2)^{n-1}\,dx = 2^{2-2n}\int_0^1 (1-x^2)^{n-1}\,dx$$

Let $x^2 = u$: $= 2^{1-2n}B\left(\frac{1}{2},n\right) = 2^{1-2n}\dfrac{\Gamma(1/2)\Gamma(n)}{\Gamma(n+1/2)} = \dfrac{\sqrt{\pi}\,2^{1-2n}\,\Gamma(n)}{\Gamma(n+1/2)}$.

Equating with $\dfrac{[\Gamma(n)]^2}{\Gamma(2n)}$:

$$2^{2n-1}\,\Gamma(n)\,\Gamma\left(n+\frac{1}{2}\right) = \sqrt{\pi}\,\Gamma(2n) \checkmark$$

Chapter 5 – Area Under Plane Curves

Cartesian & Polar curves | Standard results

Q1 **Find area enclosed by ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.**

By symmetry:

$$A = 4\int_0^a y\,dx = 4\int_0^a \frac{b}{a}\sqrt{a^2-x^2}\,dx = \frac{4b}{a}\left[\frac{x\sqrt{a^2-x^2}}{2}+\frac{a^2}{2}\sin^{-1}\frac{x}{a}\right]_0^a = \frac{4b}{a}\cdot\frac{\pi a^2}{4}$$

$A = \pi ab$

Q2 **Find area bounded by astroid $x^{2/3}+y^{2/3}=a^{2/3}$.**

Parametric: $x = a\cos^3 t$, $y = a\sin^3 t$, $t\in[0,2\pi]$. $dx = -3a\cos^2 t\sin t\,dt$.

$$A = 4\int_0^{\pi/2} y\,(-dx) = 4\int_0^{\pi/2} a\sin^3 t\cdot 3a\cos^2 t\sin t\,dt = 12a^2\int_0^{\pi/2}\sin^4 t\cos^2 t\,dt$$

By Wallis: $\int_0^{\pi/2}\sin^4 t\cos^2 t\,dt = \frac{3\cdot1\cdot1}{6\cdot4\cdot2}\cdot\frac{\pi}{2} = \frac{\pi}{32}$.

$A = 12a^2\cdot\frac{\pi}{32} = \frac{3\pi a^2}{8}$

Q3 **Find area of one loop of $r = a\cos 3\theta$ and the whole area.**

One loop: $\theta\in[-\pi/6,\pi/6]$.

$$A_{\text{loop}} = \frac{1}{2}\int_{-\pi/6}^{\pi/6} r^2\,d\theta = \int_0^{\pi/6} a^2\cos^2 3\theta\,d\theta = \frac{a^2}{2}\left[\theta+\frac{\sin 6\theta}{6}\right]_0^{\pi/6} = \frac{a^2}{2}\cdot\frac{\pi}{6} = \frac{\pi a^2}{12}$$

There are 3 petals (rose with $n=3$).

One loop $=\frac{\pi a^2}{12}$; **Total area** $= 3\times\frac{\pi a^2}{12} = \frac{\pi a^2}{4}$

Q4 **Find area of one loop of $r = a\cos 4\theta$.**

8-petal rose. One loop: $\theta \in [-\pi/8, \pi/8]$.

$$A_{\text{loop}} = \int_0^{\pi/8} a^2 \cos^2 4\theta \, d\theta = \frac{a^2}{2}\left[\theta + \frac{\sin 8\theta}{8}\right]_0^{\pi/8} = \frac{\pi a^2}{16}$$

One loop $= \dfrac{\pi a^2}{16}$; **Total (8 loops)** $= \dfrac{\pi a^2}{2}$

Q5 **Find area bounded by $a^2y^2 = x^3(2a - x)$ (Cissoid of Diocles).**

Curve exists for $0 \le x \le 2a$. Let $x = 2a\sin^2 t$, $2a - x = 2a\cos^2 t$:

$$A = \frac{2}{a}\int_0^{2a} x^{3/2}\sqrt{2a - x}\, dx = \frac{2}{a}\int_0^{\pi/2} (2a\sin^2 t)^{3/2}\sqrt{2a}\cos t \cdot 4a \sin t \cos t \, dt$$

$$= \frac{2}{a} \cdot 4a \cdot (2a)^{3/2} \cdot \sqrt{2a}\int_0^{\pi/2} \sin^4 t \cos^2 t \, dt = 32a^2 \cdot \frac{\pi}{32} = \pi a^2$$

$A = \pi a^2$

Chapter 6 – Volumes of Solids of Revolution

Disc/Washer Method | Parametric Curves

Q1 **For astroid $x^{2/3} + y^{2/3} = a^{2/3}$, show volume of solid formed by revolution about an axis $= \dfrac{32\pi a^3}{105}$.**

Revolution about the x-axis. Using parametric form $x = a\cos^3 t$, $y = a\sin^3 t$:

$$V = \pi \int_{-a}^{a} y^2\,dx = 2\pi \int_0^{\pi/2} (a\sin^3 t)^2 \cdot 3a\cos^2 t \sin t\,dt = 6\pi a^3 \int_0^{\pi/2} \sin^7 t \cos^2 t\,dt$$

By Wallis/Beta: $\int_0^{\pi/2} \sin^7 t\cos^2 t\,dt = B(4, \frac{3}{2})/2 = \dfrac{\Gamma(4)\Gamma(3/2)}{2\Gamma(11/2)} = \dfrac{6 \cdot \frac{\sqrt{\pi}}{2}}{2 \cdot \frac{945\sqrt{\pi}}{256}} \cdot \frac{1}{\cdots} = \dfrac{16}{315}$

$$V = 6\pi a^3 \cdot \frac{16}{315} = \frac{96\pi a^3}{315}$$

$V = \dfrac{32\pi a^3}{105}$ ✓

Q2 **Find volume of solid generated by revolving ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ about its axes.**

Revolution about x-axis:

$$V_x = \pi \int_{-a}^{a} y^2\,dx = 2\pi \int_0^a \frac{b^2}{a^2}(a^2 - x^2)\,dx = \frac{2\pi b^2}{a^2}\left[a^2 x - \frac{x^3}{3}\right]_0^a = \frac{2\pi b^2}{a^2} \cdot \frac{2a^3}{3}$$

$V_x = \dfrac{4}{3}\pi a b^2$

Revolution about y-axis (swap $a \leftrightarrow b$):

$V_y = \dfrac{4}{3}\pi a^2 b$

Note: When $a = b = r$ (sphere), $V = \frac{4}{3}\pi r^3$ ✓

Chapter 7 – Co-ordinate Geometry

Transformation of Axes | Reduction to Standard Form | Conics | Planes

Q1 **Find transformation equation of $3x^2 + 2xy + 3y^2 - 18x - 22y + 50 = 0$ when origin shifted to $(2, 3)$ and axes rotated 45ř.**

Step 1: Shift origin to $(2, 3)$: set $x = X + 2$, $y = Y + 3$.
After substituting and expanding (all linear terms cancel):

$$3X^2 + 2XY + 3Y^2 = 7$$

Step 2: Rotate 45ř: $X = \frac{x' - y'}{\sqrt{2}}$, $Y = \frac{x' + y'}{\sqrt{2}}$.

$$3 \cdot \frac{(x' - y')^2}{2} + 2 \cdot \frac{(x' - y')(x' + y')}{2} + 3 \cdot \frac{(x' + y')^2}{2} = 7$$

$$\frac{3}{2}(x'^2 - 2x'y' + y'^2) + (x'^2 - y'^2) + \frac{3}{2}(x'^2 + 2x'y' + y'^2) = 7$$

$$4x'^2 + 2y'^2 = 7$$

$\frac{x'^2}{7/4} + \frac{y'^2}{7/2} = 1$ **(Ellipse)**

Q2 **Determine T.E. of $11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0$ (origin unchanged, axis rotated by $\theta = \tan^{-1}(-4/3)$).**

To remove xy term: $\tan 2\theta = \frac{2h}{a - b} = \frac{24}{11 - 4} = \frac{24}{7}$.
Solving: $\cos\theta = 3/5$, $\sin\theta = 4/5$.
Rotation: $x = \frac{3x' - 4y'}{5}$, $y = \frac{4x' + 3y'}{5}$.
After substitution and simplification:

$$20x'^2 - 5y'^2 - 20x' - 40y' - 5 = 0 \implies 4x'^2 - y'^2 - 4x' - 8y' - 1 = 0$$

Complete the square: $4(x' - \frac{1}{2})^2 - (y' + 4)^2 = 1 + 1 - 16 = -14...$ (centre shifts accordingly).

$\frac{(x' - \frac{1}{2})^2}{1/4} - \frac{(y' + 4)^2}{1} = 1$ **(after completing the square) — Hyperbola**

Q3 **Reduce to standard form and find properties:**

(a) $16x^2 - 24xy + 9y^2 - 104x - 172y + 44 = 0$
$\Delta = B^2 - 4AC = 576 - 576 = 0 \Rightarrow$ **Parabola**.
Rotate: $\tan 2\theta = -24/7 \Rightarrow \cos\theta = 4/5$, $\sin\theta = 3/5$.
After rotation and completing the square:

Reduces to $Y'^2 = 20X'$ (standard parabola).

(b) $x^2 - 6xy + 9y^2 - 2x - 3y + 1 = 0$
$B^2 - 4AC = 36 - 36 = 0 \Rightarrow$ **Parabola**. $(x - 3y)^2 = 2x + 3y - 1$.

Reduces to standard parabola after rotation and translation.

(c) $x^2 + 2xy + y^2 - 6x - 2y + 4 = 0$
$B^2 - 4AC = 4 - 4 = 0 \Rightarrow$ **Parabola**. $(x + y)^2 = 6x + 2y - 4$.

Standard parabola form obtained after 45ř rotation.

(d) $8x^2 + 4xy + 5y^2 - 24x - 24y = 0$
$B^2 - 4AC = 16 - 160 = -144 < 0 \Rightarrow$ **Ellipse**.

Reduces to standard ellipse $\dfrac{X'^2}{p^2} + \dfrac{Y'^2}{q^2} = 1$ after rotation and translation.

Plane (3D Geometry)

Q1 **Find equations of planes:**

(i) Through $(1, 1, 2)$, perpendicular to each of $2x - 2y - 4z = 6$ and $3x + y + 6z = 4$.
Normal $\mathbf{n} = \mathbf{n}_1 \times \mathbf{n}_2$, where $\mathbf{n}_1 = (2, -2, -4)$, $\mathbf{n}_2 = (3, 1, 6)$:

$$\mathbf{n} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -2 & -4 \\ 3 & 1 & 6 \end{vmatrix} = \mathbf{i}(-12 + 4) - \mathbf{j}(12 + 12) + \mathbf{k}(2 + 6) = (-8, -24, 8) \parallel (-1, -3, 1)$$

Plane through $(1, 1, 2)$: $-(x - 1) - 3(y - 1) + (z - 2) = 0$

$x + 3y - z = 2$

(ii) Through intersection of $7x - 4y + 7z + 16 = 0$ and $4x + 3y - 2z + 3 = 0$, parallel to $3x - 7y + 9z + 5 = 0$.
Family: $(7x - 4y + 7z + 16) + \lambda(4x + 3y - 2z + 3) = 0$.

Normal $(7+4\lambda, -4+3\lambda, 7-2\lambda)$ parallel to $(3,-7,9)$:

$$\frac{7+4\lambda}{3} = \frac{-4+3\lambda}{-7} \Rightarrow -7(7+4\lambda) = 3(-4+3\lambda) \Rightarrow \lambda = -1$$

$3x - 7y + 9z + 13 = 0$

(iii) Through midpoint of $(2,-3,1)$ and $(3,4,-1)$, normal to line joining $(3,4,-1)$ and $(2,-1,5)$.

Midpoint $= \left(\frac{5}{2}, \frac{1}{2}, 0\right)$. Direction $(2-3, -1-4, 5+1) = (-1,-5,6)$.

$x + 5y - 6z = 5$

Q2 **Variable plane at constant distance p from origin meets axes at A, B, C. Show locus of centroid of tetrahedron $OABC$ is $x^{-2} + y^{-2} + z^{-2} = 16p^{-2}$.**

Plane: $\dfrac{x}{a} + \dfrac{y}{b} + \dfrac{z}{c} = 1$. Distance from O:

$$p = \frac{1}{\sqrt{1/a^2 + 1/b^2 + 1/c^2}} \implies \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{p^2} \quad \cdots (*)$$

Centroid $G = (a/4, b/4, c/4)$. Let $x = a/4$, $y = b/4$, $z = c/4$, so $a = 4x$, $b = 4y$, $c = 4z$. Substitute in $(*)$:

$$\frac{1}{16x^2} + \frac{1}{16y^2} + \frac{1}{16z^2} = \frac{1}{p^2}$$

$x^{-2} + y^{-2} + z^{-2} = 16p^{-2}$ ✓

Chapter 8 – Rectangular Co-ordinate (3D)

Direction Cosines | Direction Ratios | Angle Between Lines

Key Definitions

Direction Cosines $l, m, n = \cos\alpha, \cos\beta, \cos\gamma$: cosines of angles with $+x, +y, +z$ axes. $l^2 + m^2 + n^2 = 1$.

Direction Ratios Any a, b, c proportional to l, m, n: $l = \dfrac{a}{\sqrt{a^2+b^2+c^2}}$, etc.

Q1 **If l, m, n are direction cosines of a line, show $l^2 + m^2 + n^2 = 1$.**

A unit vector along the line has components $(l, m, n) = (\cos\alpha, \cos\beta, \cos\gamma)$.
Its magnitude squared $= l^2 + m^2 + n^2 = |\hat{u}|^2 = 1$. ✓

Q2 **If $P = (2, 3, 4)$ and $Q = (1, 1, -1)$, find angle between OP and OQ.**

$$\overrightarrow{OP} = (2, 3, 4),\ |\overrightarrow{OP}| = \sqrt{29}; \qquad \overrightarrow{OQ} = (1, 1, -1),\ |\overrightarrow{OQ}| = \sqrt{3}$$

$$\overrightarrow{OP} \cdot \overrightarrow{OQ} = 2 + 3 - 4 = 1; \qquad \cos\theta = \frac{1}{\sqrt{29}\cdot\sqrt{3}} = \frac{1}{\sqrt{87}}$$

$$\theta = \cos^{-1}\frac{1}{\sqrt{87}}$$

Q3 **Lines with d.c.'s connected by $al + bm + cn = 0$ and $ul^2 + vm^2 + wn^2 = 0$ are: perpendicular if $a^2(v+w) + b^2(w+u) + c^2(u+v) = 0$; parallel if $\dfrac{a^2}{u} + \dfrac{b^2}{v} + \dfrac{c^2}{w} = 0$.**

From $al + bm + cn = 0$: $l = -(bm + cn)/a$. Substituting in $ul^2 + vm^2 + wn^2 = 0$:

$$(ub^2 + va^2)m^2 + 2ubcmn + (uc^2 + wa^2)n^2 = 0$$

This is quadratic in m/n with roots m_1/n_1 and m_2/n_2:

$$\frac{m_1 m_2}{n_1 n_2} = \frac{uc^2 + wa^2}{ub^2 + va^2}; \qquad \frac{l_1 l_2}{n_1 n_2} = \text{(similarly found)}$$

Perpendicular: $l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$ reduces to $a^2(v+w) + b^2(w+u) + c^2(u+v) = 0$. ✓

Parallel: Equal roots (discriminant $= 0$) gives $a^2/u + b^2/v + c^2/w = 0$. ✓

Q4 **Prove the angle between two diagonals of a cube is $\cos^{-1}\frac{1}{3}$.**

Consider unit cube. Two body diagonals:

$$\overrightarrow{d_1} : (0,0,0) \to (1,1,1) \;\Rightarrow\; \mathbf{d_1} = (1,1,1)$$

$$\overrightarrow{d_2} : (1,0,0) \to (0,1,1) \;\Rightarrow\; \mathbf{d_2} = (-1,1,1)$$

$$\cos\theta = \frac{\mathbf{d_1}\cdot\mathbf{d_2}}{|\mathbf{d_1}||\mathbf{d_2}|} = \frac{-1+1+1}{\sqrt{3}\cdot\sqrt{3}} = \frac{1}{3}$$

$\theta = \cos^{-1}\frac{1}{3}$ ✓

End of Solution Book

All problems from the L1 T1 Suggestion Sheet have been fully solved.

Review each step, practise the standard proofs, and master the theorems.

Best of luck in your examinations!